

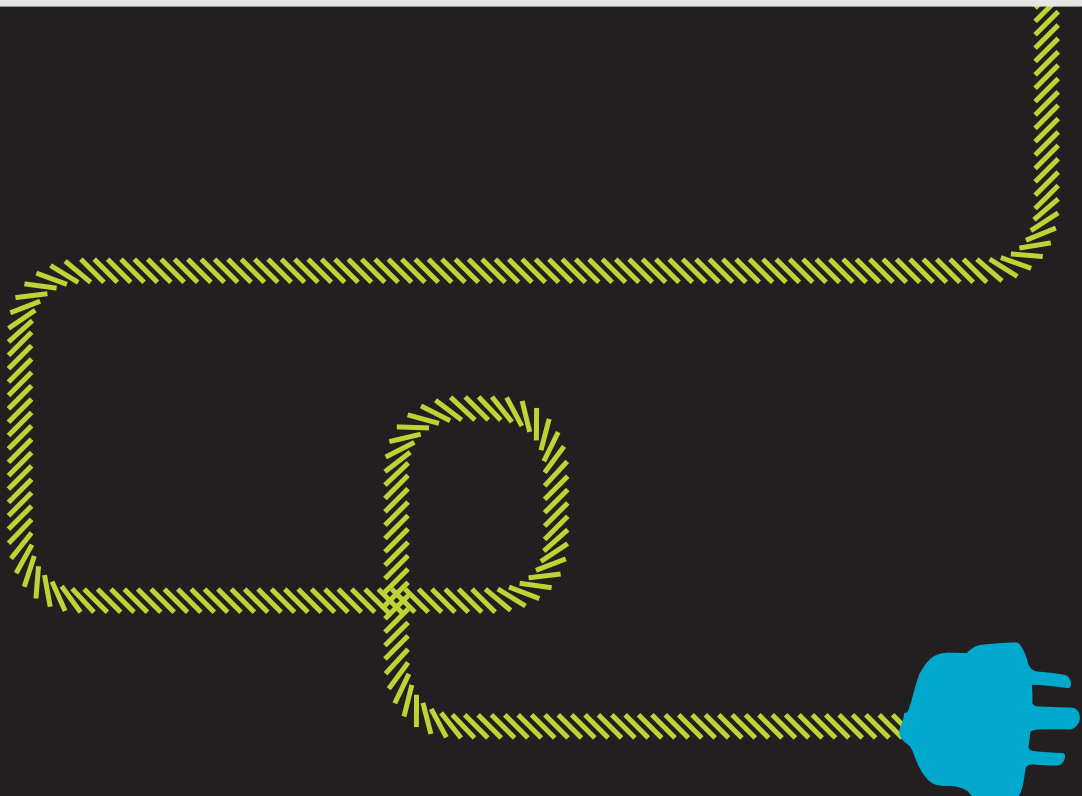
7

THE PARALLEL CIRCUIT

In a parallel circuit, all components connect directly to the applied voltage. This chapter looks at ways of determining the total resistance of any type of parallel circuit, and how to calculate current in each branch of the circuit. The chapter also describes power in the parallel circuit, and the effects of an open- or short-circuit.

Chapter outline

- 7.1 The parallel circuit
- 7.2 Voltage in the parallel circuit
- 7.3 Current in the parallel circuit
- 7.4 Resistance in the parallel circuit
- 7.5 Summary of equations
- 7.6 Power in the parallel circuit
- 7.7 Faults in the parallel circuit
- 7.8 Comparison to the series circuit



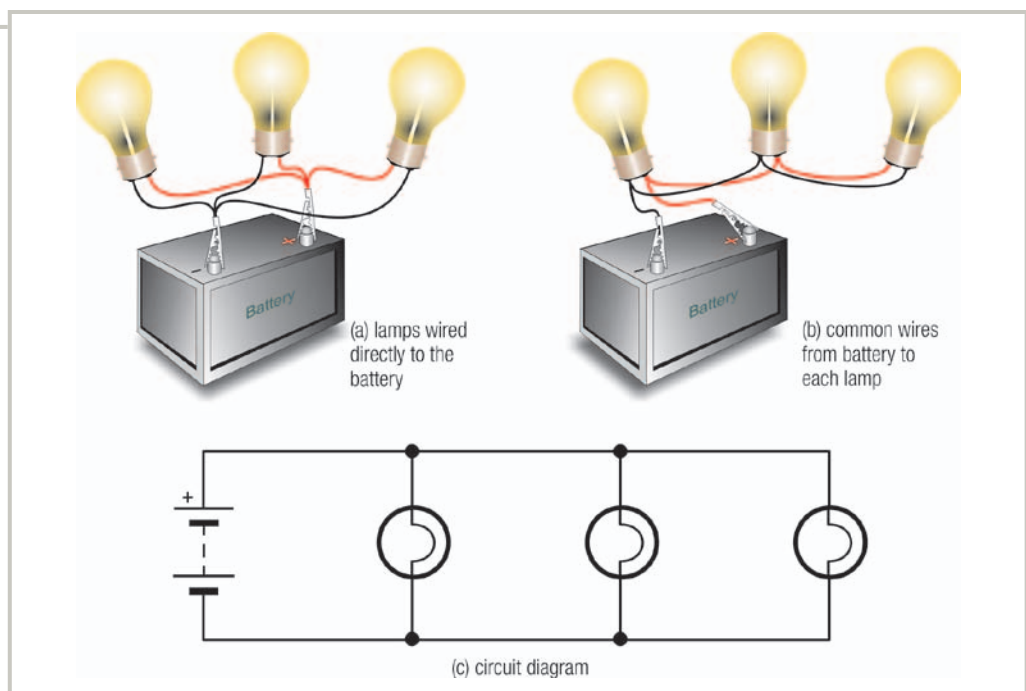
7.1 The parallel circuit

When you plug two appliances into the same power outlet, you are connecting the appliances in *parallel*. In a parallel circuit, all the components connect directly to the supply voltage. In a car, the lights and electrical accessories are all connected in parallel with the car battery – although they are individually controlled with their own on-off switches. In a house, every appliance is connected in parallel with the mains supply.

Figure 7.1 shows three lamps connected in parallel with a battery. There are several ways the wires can be run to do this, such as wiring each lamp directly to the battery as in (a). A more typical way is with a common set of wires that run the length of the circuit, as shown in (b). Regardless of how the wiring is arranged, the circuit diagram is the same.

FIGURE 7.1

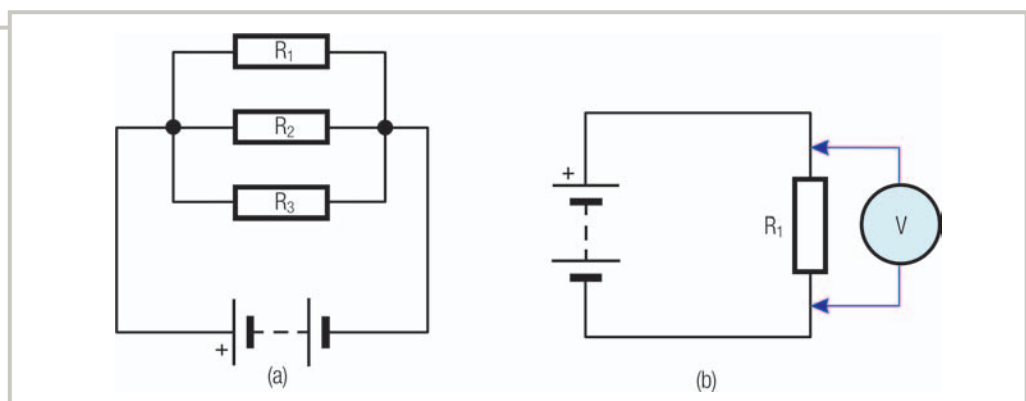
Three lamps connected in parallel with different ways of running the wiring



A parallel circuit can be drawn in a number of ways. In Figure 7.2(a), three resistors are connected in parallel with a battery. Regardless of how the parallel circuit is drawn, each component connects directly to the battery. Circuit (b) shows a single resistor connected to a battery, with a voltmeter connected across, or in parallel with, the resistor. As we've explained, a voltmeter is always connected in parallel with a component to measure the voltage across that component.

FIGURE 7.2

More examples of the parallel connection



Unlike the series circuit, a component in a parallel circuit, therefore, might have more than two conductors connected to its terminals. For example, in Figure 7.1(b) the only component with two conductors is the lamp on the right.

7.2 Voltage in the parallel circuit

You can already see that because each component in a parallel circuit connects directly to the voltage source, the voltage across each component must be the same as the voltage of the voltage source.

This is shown in Figure 7.3 in which three lamps are connected in parallel with a 12 V battery. As the diagram shows, the voltmeter will read 12 V if it's connected across any of the three lamps, or across the battery.

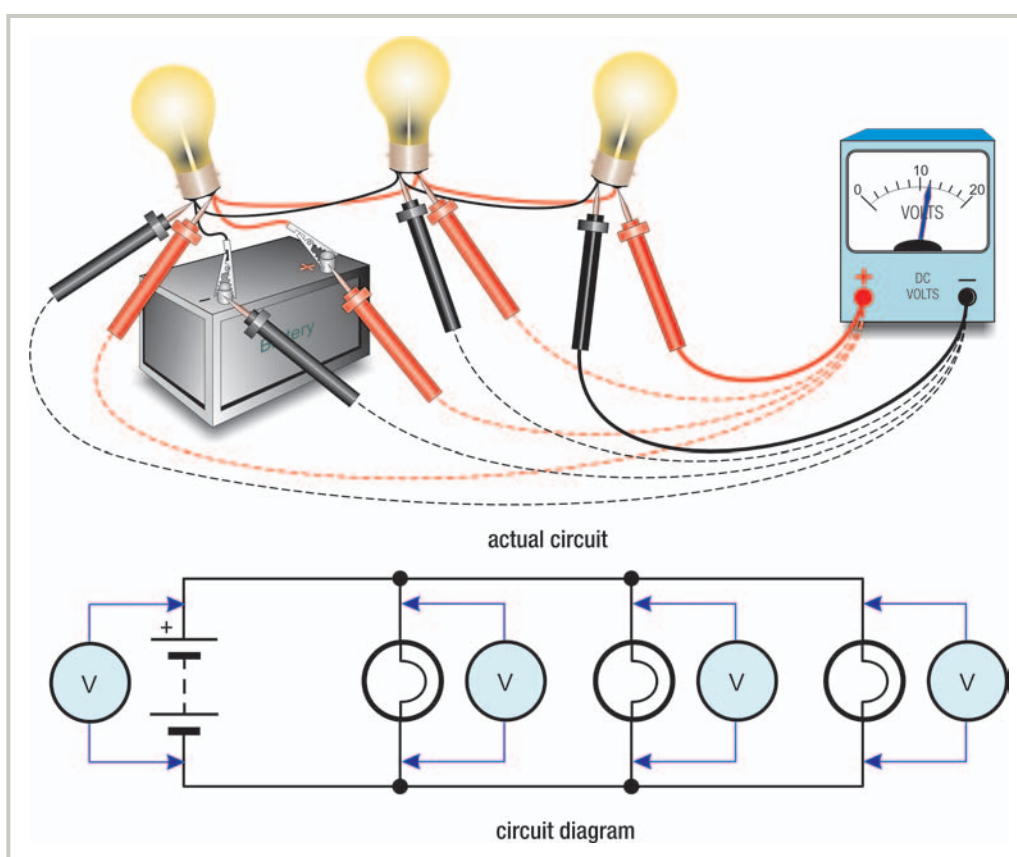


FIGURE 7.3

The voltage across each component in a parallel circuit is equal to the source voltage

The circuit diagram in Figure 7.3 shows the same thing. Here, voltages V_1 , V_2 and V_3 are all the same, and equal to the applied voltage (V). This is different to the series circuit, where voltage drops across each component depending on the value of its resistance and the current flowing in the circuit.

SUMMING UP...

- The voltage is the same across all components in a parallel circuit, and equals the applied voltage.

7.3 Current in the parallel circuit

Unlike the series circuit, which has one path for the current, a parallel circuit has a number of paths. As well, the current in each path (or branch) is usually different. This is shown in Figure 7.4 where three resistors (with different values) are connected in parallel with a 10 V supply. An ammeter is connected to measure the current in each resistor and another ammeter is connected to show the total current taken from the supply.

FIGURE 7.4

Measuring current in a parallel circuit

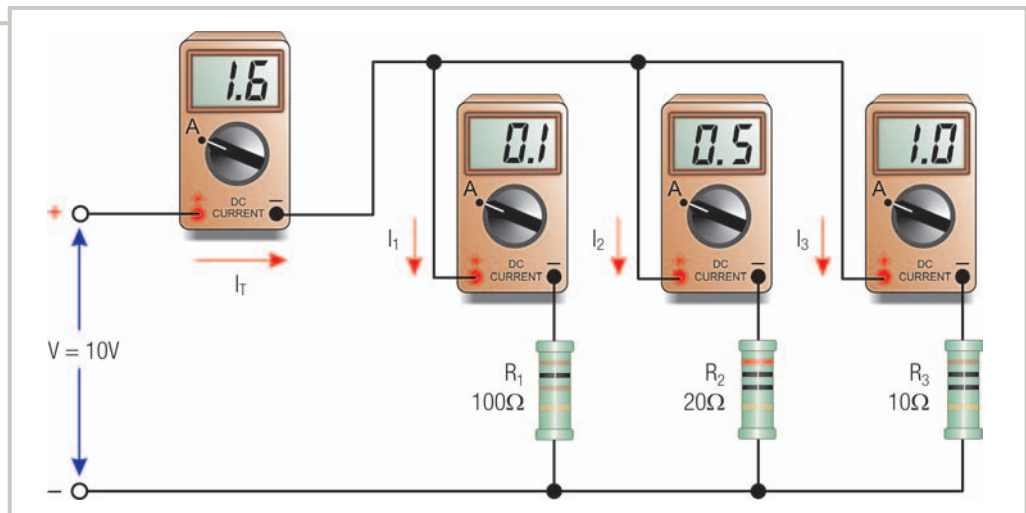
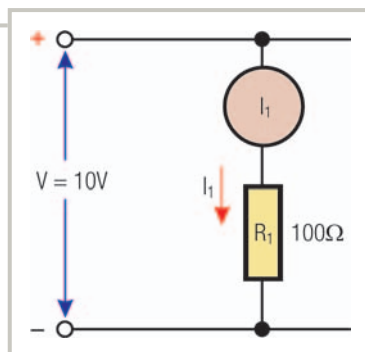


FIGURE 7.5

Looking at R_1 only



The values of each branch current can be found with Ohm's law by looking at each branch separately. The circuit in Figure 7.5 shows the conditions for resistor R_1 only.

$$I_1 = \frac{V}{R_1} = \frac{10}{100} = 0.1 \text{ A}$$

Notice that this is the value indicated by the ammeter for I_1 in Figure 7.4. The same method is used to find the current in R_2 . Figure 7.6 shows the circuit for R_2 only, where I_2 is also found with Ohm's law.

$$I_2 = \frac{V}{R_2} = \frac{10}{20} = 0.5 \text{ A}$$

The current in R_3 is also found with Ohm's law.

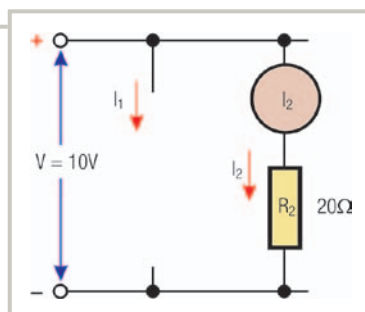
$$I_3 = \frac{V}{R_3} = \frac{10}{10} = 1 \text{ A}$$

When calculating a branch current, each branch is treated on its own, as if the other branches don't exist. This applies no matter how many branches there are in a parallel circuit.

If you compare the resistance values in each branch and the current in that branch, you can see that the higher the branch resistance, the lower the branch current. In other words, the current in a branch of a parallel circuit is *inversely* proportional to the resistance of the branch. This is simply an application of Ohm's law, which says that current is inversely proportional to resistance.

FIGURE 7.6

Looking at R_2 only



The next thing to consider is the total current taken from the supply. Figure 7.7 is the circuit diagram of Figure 7.4, and shows that the branch currents flow from the positive potential of the supply, through each resistor then back to the supply. Because each branch current comes from the supply, the total current (I_T) taken from the supply is therefore the sum of the individual branch currents.

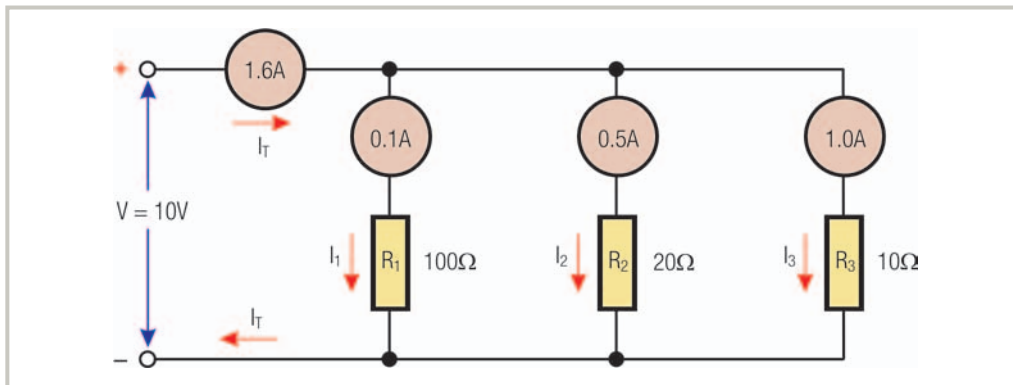


FIGURE 7.7

The total current taken from the supply equals the sum of the branch currents

In mathematical terms, the total current (I_T) in Figure 7.7 equals $I_1 + I_2 + I_3$. If more parallel components are added to the circuit, the total current will increase. Therefore, in a parallel circuit the total current equals the sum of *all* the branch currents.

SUMMING UP...

- A branch current equals the applied voltage divided by the resistance of the branch. That is:

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3}, \dots \text{ where: } I_1, I_2, I_3 \text{ (etc.) are the branch currents in amperes.}$$

- V is the applied voltage in volts.
- R_1, R_2, R_3 (etc.) are the resistance values (in ohms) of each branch.
- Branch current is inversely proportional to the resistance of the branch.
- The total current equals the sum of all the branch currents. That is:
 $I_T = I_1 + I_2 + I_3 + \dots$ where: I_T is the total current in amperes taken from the supply.

Kirchhoff's current law

This equation to find current in a parallel circuit is similar to that for voltage drops in a series circuit, where the sum of the voltage drops equals the applied voltage (Kirchhoff's voltage law). It makes sense to have a similar law for parallel circuits, this time for current. When Kirchhoff came up with his voltage law, he also developed a law for current which says:

The algebraic sum of all the currents entering a junction equals the algebraic sum of the currents leaving the junction.

Figure 7.8 shows how the currents divide for the circuit of Figure 7.7. The total circuit current (I_T) enters junction 1, and two currents leave the junction: I_1 which flows through R_1 , and $I_2 + I_3$ which flows into junction 2.

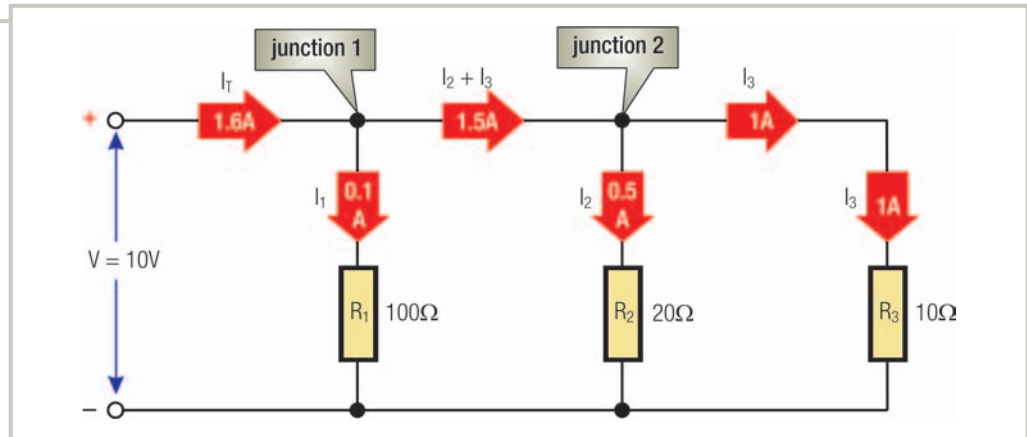


**Kirchhoff's
current law**

Notice that 1.6 A flows into and out of junction 1. There is 1.5 A entering junction 2, and 1.5 A leaves it, divided into two paths: I_2 and I_3 .

FIGURE 7.8

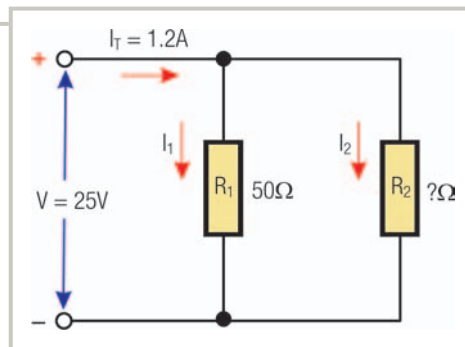
The current entering a junction equals the sum of the currents leaving the junction



The following example uses Ohm's law and Kirchhoff's current law to find the branch currents in a parallel circuit. Ohm's law is used to find I_1 . Kirchhoff's current law is used to find I_2 . (The equation has to be transposed in terms of I_2 .)

EXAMPLE 7.1

Calculate the currents flowing in resistors R_1 and R_2 in Figure 7.9.

FIGURE 7.9**Solution**

Values $R_1 = 50 \Omega$
 $I_T = 1.2 \text{ A}$
 $V = 25 \text{ V}$

To find I_1

Equation $I_1 = \frac{V}{R_1} = \frac{25}{50}$

Answer $I_1 = 0.5 \text{ A}$

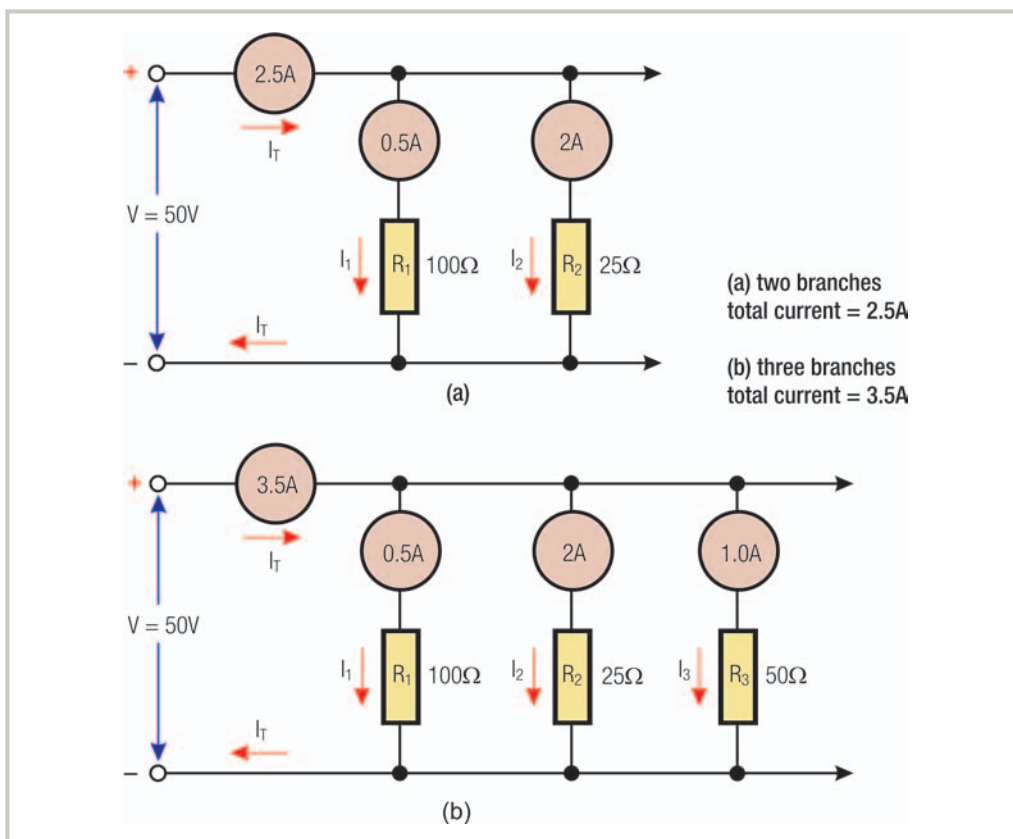
To find I_2

Equation $I_2 = I_T - I_1 = 1.2 - 0.5$

Answer $I_2 = 0.7 \text{ A}$

7.4 Resistance in the parallel circuit

We've shown that a parallel circuit has a number of branches, with each branch taking current from the voltage supply. The more branches, the more current. For example, Figure 7.10(a) has two branches, each taking a particular value of current. If a branch is added, as in (b) there will be more current taken. That is, adding another resistor in parallel will cause the total current to increase, which means the total resistance of the circuit must have decreased.

**FIGURE 7.10**

Adding another resistor in parallel increases the total current

The total resistance (R_T) of Figure 7.10(a) and (b) can be found with Ohm's law, as the applied voltage (V) and the total current (I_T) are known. That is:

$$R_T = \frac{V}{I_T} = \frac{50}{2.5} = 20\ \Omega$$

$$R_T = \frac{V}{I_T} = \frac{50}{3.5} = 14.29\ \Omega$$

As the answers show, adding another resistive branch to a parallel circuit always reduces the total resistance of the circuit, and always increases the total circuit current. On the other hand, removing a branch increases the total resistance, and reduces the total current.

Resistance equation

When you don't know the total current or the applied voltage, the total resistance of a parallel circuit must be found in other ways. The following shows the derivation of an equation to find the total resistance (R_T) from the individual resistance values. It starts with Ohm's law, which can be used with any circuit if you know the applied voltage and the total current.

$$1 \quad R_T = \frac{V}{I_T} \text{ where } V \text{ is the applied voltage and } I_T \text{ is the total circuit current}$$

This equation can be transposed in terms of I_T giving,

$$2 \quad I_T = \frac{V}{R_T}$$

The total current in a parallel circuit equals the sum of the branch currents. Each branch current can be determined by Ohm's law, as in Example 7.1. Therefore, the total current can be found by adding the Ohm's law equations for each branch current. That is,

$$3 \quad I_T = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} + \dots$$

Notice that equations 2 and 3 both equal I_T . Therefore, the right-hand side of these equations must equal each other, giving equation 4.

$$4 \quad \frac{V}{R_T} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} + \dots$$

Because V is common to all parts of equation 4, its value doesn't matter as it affects both sides equally. Therefore, by dividing both sides by V , you get a *general* equation to find the total resistance of a parallel circuit. This equation is:

$$5 \quad \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Equation 5 gives the reciprocal of the total resistance. In other words, we can say that the reciprocal of the total resistance of a parallel circuit equals the sum of the reciprocals of each branch resistance. Equation 6 below gives the total resistance (R_T) of a parallel circuit, and is derived from equation 5.

$$6 \quad R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots}$$

As a useful check, the total resistance of a parallel circuit *must* be lower than the lowest value branch resistance.

**Resistance
of a parallel
circuit**



SUMMING UP...

- The resistance of any parallel circuit: $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$
- The total resistance of a parallel circuit is always less than the smallest value of branch resistance.

Example 7.2 shows how to use this equation, with suggestions on using a calculator given in Figure 7.12.

EXAMPLE 7.2

Calculate the total resistance of the circuit in Figure 7.11.

Solution

Values $R_1 = 100 \, \Omega$, $R_2 = 220 \, \Omega$, $R_3 = 330 \, \Omega$
 $R_T = ?$





Equation
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Working
$$\frac{1}{R_T} = \frac{1}{100} + \frac{1}{220} + \frac{1}{330}$$

(add the reciprocals of each resistance)

$$\frac{1}{R_T} = 0.017575757$$

(the sum of the reciprocals)

Answer $R_T = 56.89655172$ or 56.9Ω (rounded off to two decimal places)

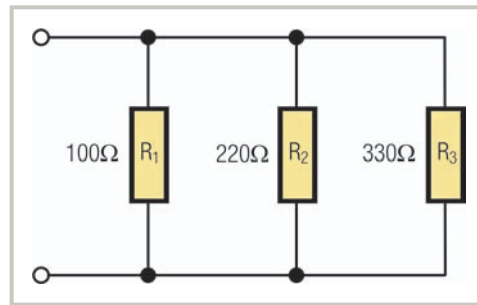


FIGURE 7.11

reciprocal key could be $1/x$ inv x^{-1}
example uses x^{-1}

1 0 0 x^{-1} (to enter $1/100$)
+
2 2 0 x^{-1} (to enter $1/220$)
+
3 3 0 x^{-1} (to enter $1/330$)
= (for sum of the reciprocals)
 x^{-1} = (for final answer of 56.89655172)

FIGURE 7.12

Calculator key strokes to solve Example 7.2

The next example has resistor values expressed with engineering notation. These values are entered into a calculator as described in Chapter 3. Otherwise, the sequence is the same as for the last example.

EXAMPLE 7.3

What is the total resistance of the circuit in Figure 7.13?

Solution

Values $R_1 = 8 \text{ k}\Omega$
 $R_2 = 2 \text{ k}\Omega$
 $R_3 = 1.6 \text{ k}\Omega$
 $R_T = ?$

Equation
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_T} = \frac{1}{8 \times 10^3} + \frac{1}{2 \times 10^3} + \frac{1}{1.6 \times 10^3}$$

Answer $R_T = 800 \Omega$

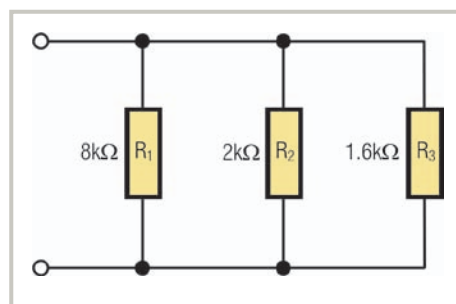


FIGURE 7.13

Two resistors in parallel

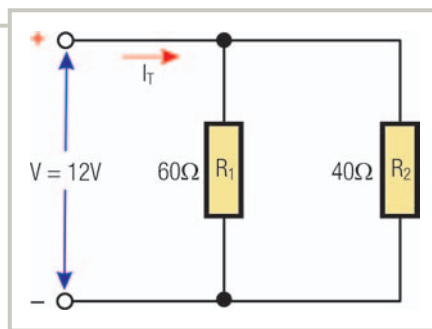
In many cases, there are only two resistors in parallel. The general equation can be used to find their combined resistance, but a simpler way is to use an equation derived from the general equation. This equation (shown below) works out to be the *product of the two resistance values divided by their sum*.

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2} \text{ where } R_1 \text{ and } R_2 \text{ are two resistors connected in parallel.}$$

EXAMPLE 7.4

Find the total resistance R_T and the total current I_T for the circuit in Figure 7.14.

FIGURE 7.14



Solution

Values $V_T = 12\text{ V}$
 $R_1 = 60\Omega$, $R_2 = 40\Omega$
 I_T and $R_T = ?$

To find R_T

Equation
$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

$$\frac{60 \times 40}{60 + 40} = \frac{2400}{100}$$

Answer $R_T = 24\Omega$

To find I_T

Equation
$$I_T = \frac{V}{R_T} = \frac{12}{24}$$

Answer $I_T = 0.5\text{ A}$

Resistors of the same value

Another special case is when all the resistors in the circuit have the same value. The equation is:

$$R_T = \frac{R}{n}$$

where

R is the value of each resistance

n is the number of resistances in the circuit.

EXAMPLE 7.5

Find the resistance of a circuit that has six 12Ω resistors connected in parallel.

Solution

Values $R = 12\Omega$
 $n = 6$

Equation
$$R_T = \frac{R}{n} = \frac{12}{6}$$

Answer $R_T = 2\Omega$

7.5 Summary of equations

Like the series circuit, current, voltage and resistance values in the parallel circuit can be found in a number of ways, depending on what you know about the circuit. Ohm's law and Kirchhoff's current law can be used. When calculating resistance you'll often use the parallel resistance equations. Here's a summary of these equations and some hints on using them.

Resistance

The total resistance of a parallel circuit is *always* less than the smallest branch resistance and gets smaller when other branches are added to the circuit.

For more than two resistors in the circuit, use equation 1 to find the total resistance. For two resistors, use equation 2. If all the resistance values are the same, use equation 3.

$$1 \quad \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \text{ (general equation)}$$

When all the resistor values have the *same* multiplier (k or M) it's often easier to enter each resistance value into a calculator without the multiplier. The multiplier is then brought back with the answer. For example, to find the total parallel resistance of a 1.2 k Ω resistor, a 1.5 k Ω resistor and a 1.8 k Ω resistor, because each resistor has the same multiplier (k), enter the values as 1.2, 1.5 and 1.8. Find the result (equals 0.486486486) then bring back the k, giving an answer (rounded off) of 0.487 k Ω . As 0.487 k is 0.487×10^3 , the answer is 487 Ω .

$$2 \quad R_T = \frac{R_1 R_2}{R_1 + R_2} \text{ (for two parallel connected resistors)}$$

As in equation 1, if the two resistors have the same multiplier, calculate the total resistance without the multipliers and bring it back with the answer. For instance, if the resistor values are 2 k Ω and 8 k Ω you get 16 (product) divided by 10 (sum), giving 1.6. Bringing back the k gives a total resistance of 1.6 k Ω .

$$3 \quad R_T = \frac{R}{n} \text{ (same value resistors, } R = \text{resistor value, } n = \text{number of resistors)}$$

Ohm's law can always be used to find the total resistance of any circuit if the applied voltage (V) and the total circuit current (I_T) are known. That is,

$$4 \quad R_T = \frac{V}{I_T} \text{ (Ohm's law)}$$

The resistance of a branch in a parallel circuit is found with Ohm's law, where V is the applied voltage and I_1 , I_2 and so on are the currents in the particular branch.

$$5 \quad R_1 = \frac{V}{I_1}, R_2 = \frac{V}{I_2}, R_3 = \frac{V}{I_3} \text{ and so on}$$

Voltage

The voltage (V) in a parallel circuit is *the same* across all components in the circuit. Voltage is found with either of the next two Ohm's law equations. Use equation 6 if you know the total current and the total resistance of the circuit.

$$6 \quad V = I_T \times R_T$$

If you know the current flowing in any *one* resistance in the circuit and you also know the value of that resistance, use equation 7 to find the applied voltage.

$$7 \quad V = I_1 \times R_1, V = I_2 \times R_2 \text{ and so on}$$



The total resistance of a parallel circuit gets smaller when another branch is connected



The voltage across any component in a parallel circuit equals the applied voltage

Current

Depending on what you know about the circuit, you can use both Kirchhoff's current law and Ohm's law to find either the total current or a branch current. Equations 7 and 8 are to find the total current (I_T).

$$8 \quad I_T = \frac{V}{R_T} \quad (\text{Ohm's law})$$

$$9 \quad I_T = I_1 + I_2 + I_3 + \dots \quad (\text{Kirchhoff's current law})$$

To find a branch current with Kirchhoff's current law, transpose equation 9 in terms of the unknown current (I). That is:

$$10 \quad I = I_T - (\text{sum of all the other branch currents in the circuit})$$

Remember that Kirchhoff's current law says that the current flowing into a junction equals the current flowing out of the junction. Use this law to find an unknown current flowing into or out of a junction if you know the other currents flowing into or out of that junction.

A branch current can also be found with Ohm's law, where:

$$11 \quad I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2} \text{ and so on}$$

Because the voltage is the same across each branch it follows that the lower the branch resistance, the higher the branch current.

Current into
a junction
equals the
current out

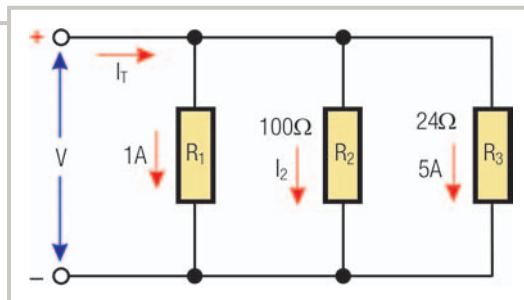


The lower
the branch
resistance,
the higher its
current



EXERCISE 7.1

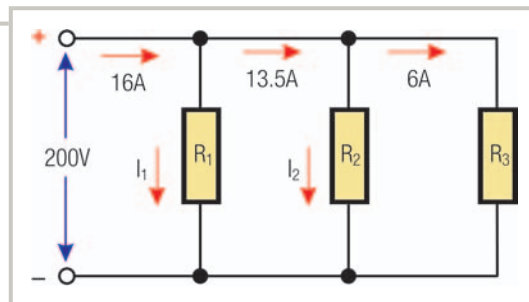
FIGURE 7.15



1 In Figure 7.15, find:

- a supply voltage V
- b current in resistor R_2
- c total current I_T
- d total resistance R_T
- e resistance value of R_1

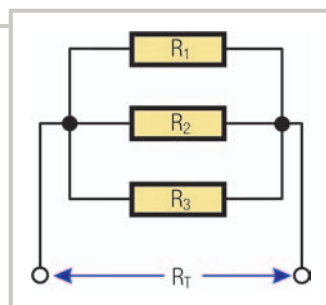
FIGURE 7.16



2 In Figure 7.16, find:

- a resistance value of R_1
- b current in resistor R_1
- c current in resistor R_2
- d total resistance R_T
- e resistance value of R_2
- f resistance value of R_3

FIGURE 7.17



3 For the circuit in Figure 7.16 find the total resistance when:

- a $R_1 = 48 \Omega$, $R_2 = 120 \Omega$, $R_3 = 72 \Omega$
- b $R_1 = 400 \Omega$, $R_2 = 630 \Omega$, $R_3 = 550 \Omega$
- c $R_1 = 4.7 \text{ k}\Omega$, $R_2 = 6.8 \text{ k}\Omega$, $R_3 = 5.6 \text{ k}\Omega$
- d all resistors = 24Ω

7.6 Power in the parallel circuit

Like the series circuit, each component in a parallel circuit takes power from the supply. But unlike a series circuit, each component is independent of the others. So when looking at the power taken by a branch, ignore the other branches and treat the branch as a single component connected to the supply. However, to find the total power taken by the circuit, you need to include all the branches.

As for the series circuit, the power taken by either the whole circuit or a part of the circuit can be found with any of the three power equations given in Chapter 4. The information you have about the circuit determines which equation to use, as shown in the following example.

EXAMPLE 7.6

Calculate the power taken by each appliance in Figure 7.18.

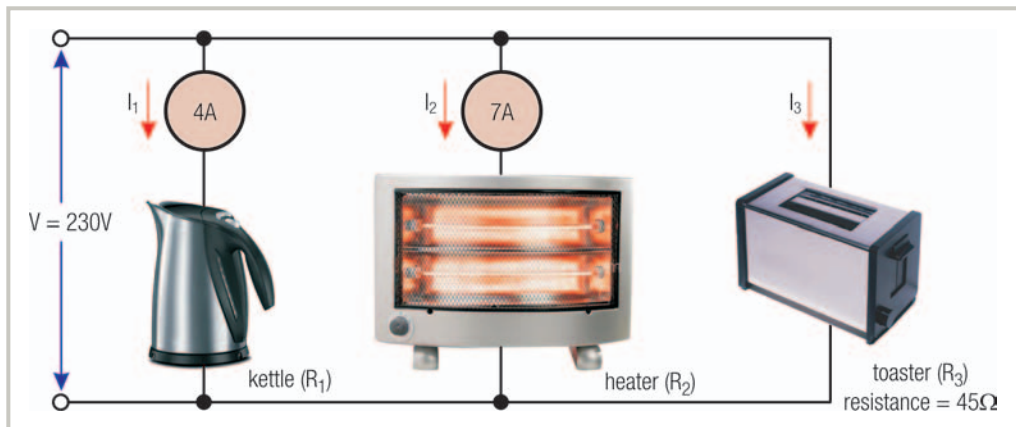


FIGURE 7.18

Solution

Values	$V = 230 \text{ V}$
	$R_3 = 45 \Omega$ (toaster)
	$I_1 = 4 \text{ A}, I_2 = 7 \text{ A}$
	$P_1, P_2, P_3, P_T = ?$
(kettle)	$P_1 = V \times I_1 = 230 \times 4$
Answer	$P_1 = 920 \text{ W}$
(heater)	$P_2 = V \times I_2 = 230 \times 7$
Answer	$P_2 = 1610 \text{ W or } 1.61 \text{ kW}$
(toaster)	$P_3 = \frac{V^2}{R_3} = \frac{240^2}{45}$
Answer	$P_3 = 1280 \text{ W or } 1.28 \text{ kW}$

Like the series circuit, the total power equals the sum of the powers taken by each branch. That is,

$$P_T = P_1 + P_2 + P_3$$

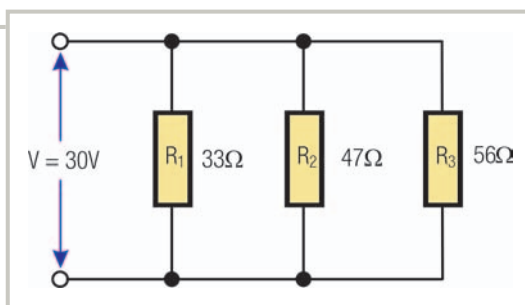
In Example 7.6, the power taken from the supply is $920 + 1610 + 1280$, which gives a total power consumption of 3810 W or 3.81 kW .

If the values of the supply voltage and individual branch resistances are known, and you want to determine the power taken by the circuit, two methods can be used. The first is to calculate the power taken by each branch, then add these to get the total power. A simpler way is to calculate the total resistance of the circuit, then use this value to calculate the total power, as shown in Example 7.7.

EXAMPLE 7.7

Calculate the total power taken by the circuit in Figure 7.19.

FIGURE 7.19



Solution

Values $V = 30 \text{ V}$
 $R_1 = 33 \Omega$
 $R_2 = 47 \Omega$
 $R_3 = 56 \Omega$
 $P_T = ?$

To find R_T

$$\text{Equation } \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\text{Working } \frac{1}{R_{eq}} = \frac{1}{33} + \frac{1}{47} + \frac{1}{56}$$

$$R_T = 14.4 \Omega$$

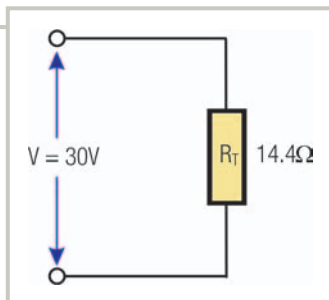
To find total power

$$\text{Equation } P_T = \frac{V^2}{R_{eq}} = \frac{30^2}{14.4}$$

$$\text{Answer } P_T = 62.5 \text{ W}$$

FIGURE 7.20

Equivalent circuit of Figure 7.19



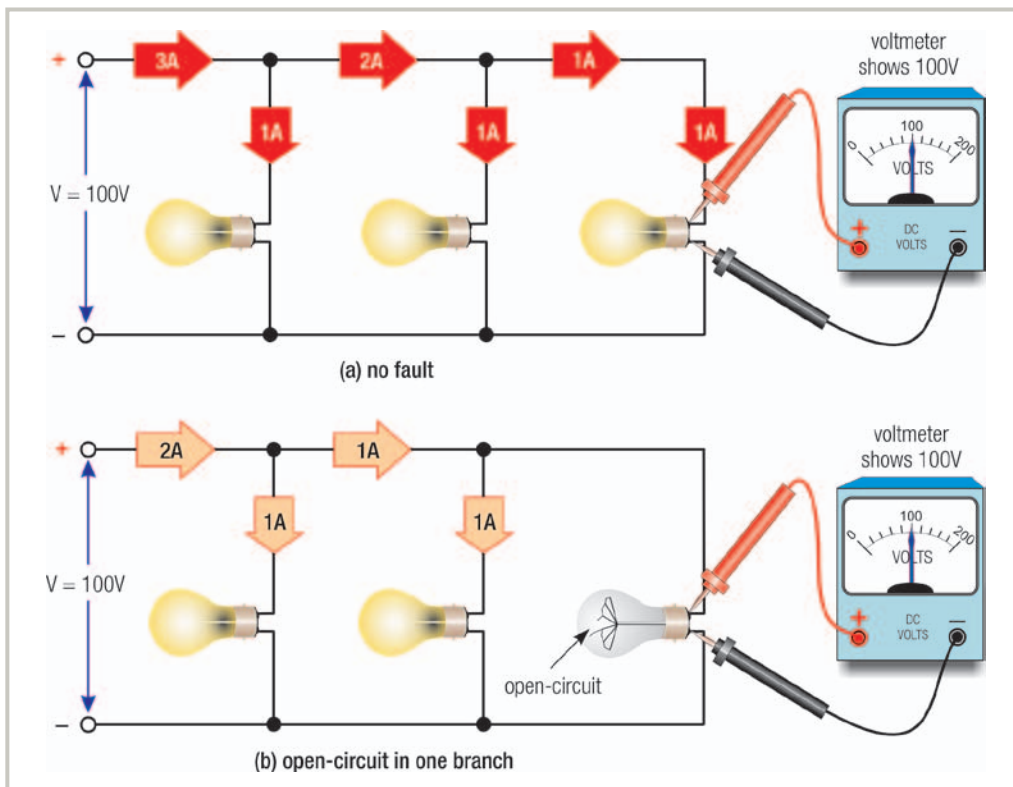
Because each branch in a parallel circuit is considered separately, if you know the power taken by a branch you can find the:

- applied voltage if you know the branch current
- branch current if you know the applied voltage.

7.7 Faults in the parallel circuit

As with any circuit, the two basic faults in a parallel circuit are an open-circuit and a short-circuit. Because each branch of a parallel circuit is independent of the others (although they all share the same supply voltage), an open-circuit in one branch won't affect the others.

This is shown in Figure 7.21(a) and (b), which shows a parallel circuit with three lamps, (a) with all lamps operating and (b) with one lamp open-circuited. Notice that there is no current in this branch and that the total current has dropped from 3 A to 2 A . However, as the diagrams show, the voltmeter across the lamp reads the supply voltage in both cases.

**FIGURE 7.21**

An open-circuit in one component of a parallel circuit causes the total current to drop

When there's an open-circuit in a branch of a parallel circuit, the:

- total power taken from the supply will drop (by the value normally taken by the branch that is now open-circuit)
- total current will drop (by the value normally taken by that branch)
- open-circuit component will no longer work, but all others in the circuit are not affected. As shown in Figure 7.21, only the open-circuited lamp goes out; in a series circuit *all* the lamps would go out
- voltage across all components will still equal the supply voltage.

Obviously, an open-circuit in a *conductor* between junctions will have a different effect. For instance, all branches after the open-circuit will be isolated from the supply. The voltage across these branches will therefore be zero. However, the voltage across the open-circuit will equal the supply voltage.

A short-circuit in a parallel circuit means there's a short-circuit across the supply voltage. This will cause a large current to flow, limited only by the capacity of the supply source and the total resistance of the circuit conductors and the short-circuit itself. Usually the circuit protection device (fuse or circuit breaker) will operate and isolate the circuit.

If the circuit is not protected, the conductors will most likely burn out. Therefore, a short-circuit across a component in a parallel circuit causes:

- the total circuit resistance to drop to a very low value
- the total current to increase by a large amount
- the voltage drop across all branches to drop to almost zero
- all components to stop working, as there's no longer any supply voltage.

7.8 Comparison to the series circuit

A parallel circuit has components that are all directly connected to the voltage source. Because each component is independent of the others, there can be any number of components in the circuit, limited by the ability of the supply source to power the components and of the circuit conductors to carry the current.

In a series circuit, the limit on the number of components is determined by the value of the supply voltage. Adding an extra component in a series circuit causes an additional voltage drop across the component, reducing the available voltage to the rest of the components. However, unlike the series circuit where a single switch controls all the components, each component in a parallel circuit can have its own on-off switch.

The main differences between the series and the parallel circuit are given in Table 7.1.

TABLE 7.1

COMPARISON OF SERIES AND PARALLEL CIRCUITS		
Function	Series circuit	Parallel circuit
Voltage	unequal	equal
Current	equal	unequal
Adding resistance	total resistance increases	total resistance decreases
Open-circuit in one component	current falls to zero in all components	current falls to zero in that component only
Short-circuit in one component	current increases, to a limit set by the other components in the circuit	current increases, limited by the resistance of the conductors and the current the power source can supply

CHAPTER SUMMARY

In a parallel circuit:

- voltage is the same across all components
- total resistance (R_T) is found with the equations:
 - $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$ (general equation)
 - $R_T = \frac{R_1 R_2}{R_1 + R_2}$ (for two parallel-connected resistors)
 - $R_T = \frac{R}{n}$ (same value resistors, R = resistor value, n = number of resistors)
 - $R_T = \frac{V}{I_T}$ (Ohm's law)
- total current taken from the supply is found with the equations:
 - $I_T = \frac{V}{R_T}$ (Ohm's law)
 - $I_T = I_1 + I_2 + I_3 + \dots$ (Kirchhoff's current law)
- Ohm's law is used to find unknown voltage, resistance and current values in any branch of the circuit
- total power (P_T) taken equals the supply voltage times the total current ($P_T = VI_T$)
- total power can also be found with the equations $P_T = I_T^2 R_T$, or $P_T = \frac{V^2}{R_T}$
- the sum of the power taken by each component equals the total power
- an open-circuit in a branch prevents only that component from receiving power
- a short-circuit will cause a considerable increase in the circuit current and cause protection devices to operate (if present).

SELF-CHECK

Answer these questions then check your answers at the back of the book.

- 1 A junction has three wires, one from the supply source, the other two feeding individual branches. If the total current is 12 A and one branch current is 7.5 A, what is the other branch current?
- 2 A parallel circuit has three branches connected to a 230 V supply. The circuit is rated to carry no more than a total current of 10 A. Calculate:
 - a the total maximum power of the circuit
 - b the minimum value of total resistance that can be connected to the circuit
 - c the current in a branch supplying a 45 Ω load.
- 3 A 120 V circuit has five appliances connected in parallel. When all appliances are operating, the total current is 5 A. Find:
 - a the total resistance of the circuit
 - b the total power dissipated
 - c total current if an appliance with a resistance of 80 ohms is removed from the circuit (switched off)
- 4 What is the total resistance of six parallel connected 100 Ω resistors?
- 5 Two resistors of 240 and 400 ohms are connected in parallel. What is their total resistance?
- 6 For the circuit in Figure 7.22, calculate:
 - a supply voltage V
 - b current in R_2
 - c current in R_3
 - d resistance of R_3
 - e total power dissipated by the circuit

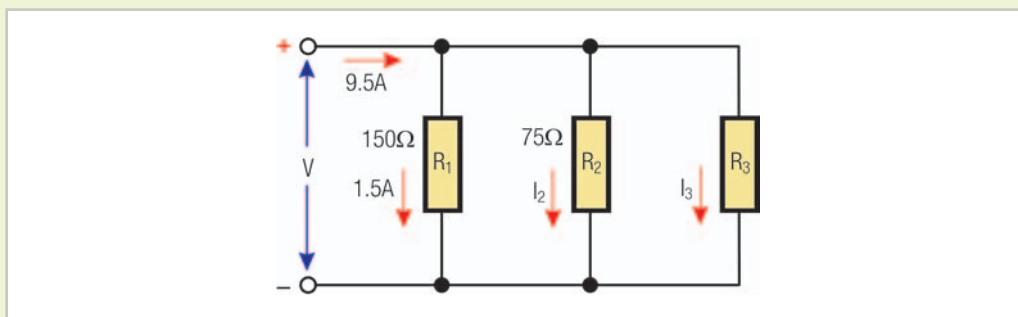


FIGURE 7.22

- 7 If eight identical lamps connected in parallel have a total resistance of 240 ohms, what is the resistance of one lamp?
- 8 For the circuit in Figure 7.23, calculate:
 - a current in R_2
 - b current in R_1
 - c values of R_1 and R_3
 - d total resistance of the circuit
 - e total power dissipated by the circuit

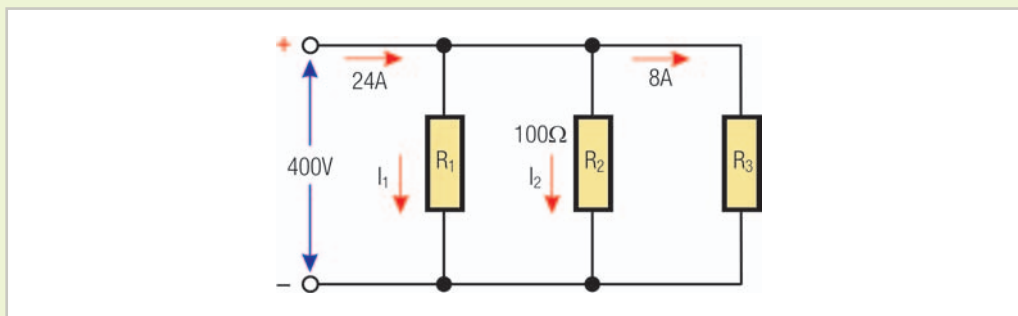
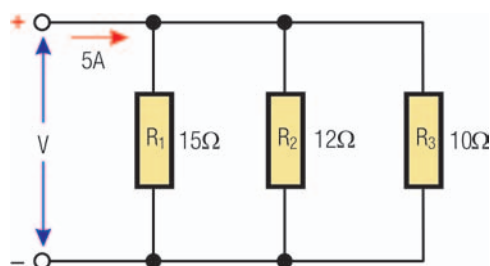


FIGURE 7.23

- 9 If resistor R_3 in Figure 7.23 is open-circuit, what is the total current (I_T) and the total power (P_T) taken by the circuit?
- 10 For the circuit in Figure 7.24, find:
- a total resistance
 - b applied voltage (V)
 - c power taken by the circuit when R_2 is removed

FIGURE 7.24



ONLINE RESOURCES

For more information about the contents of this chapter, the best search term is 'parallel circuit'. Look for websites with a .edu in the address, as these usually give uncomplicated descriptions.